

NICOLA FITZ-SIMON

Mobile: +353 833478851

Email: Nicola.fitzsimon@gmail.com

[Google scholar profile](#) Twitter: [@NicolaFitz](#)

PROFESSIONAL PROFILE

I am an applied statistician/data scientist, and my research interest is the estimation of causal effects using observational and experimental data. I have extensive experience of collaborating with clinicians, scientists, and other subject experts on observational and experimental studies to investigate important scientific questions, largely but not exclusively in the broad area of human health. I am experienced in teaching/training and communication in data science/statistics.

Research Interests:

- challenges posed for answering research questions using different kinds of data, including routinely collected big data, and data from designed studies; eg understanding impact of the data collection, measurement and missingness processes, especially for fairness & robustness
- Counterfactual reasoning to identify estimands, hence estimate causal effects in observational and experimental studies; using subject-matter knowledge and graphical methods to elucidate relationships between variables and identify control variables; statistical estimation methods for causal effects; importance of causal considerations for robust prediction
- strategies for dealing with potential biases, for instance biases due to missing data/dropouts via imputation and weighting; appropriate sensitivity analyses for assumptions

Statistical skills and experience:

- collaborating with clinicians, scientists, and other experts to develop research questions, study design, data collection, data analysis, and interpretation and presentation of results
- expertise in a broad range of both traditional statistical modelling (eg regression analyses, hierarchical models, survival models) and machine learning methods for prediction and estimation; appropriate estimation of uncertainty; frequentist and Bayesian approaches
- expert coding in statistical software, primarily R/RStudio, including Shiny, R Markdown, tidyverse; also Python, SAS, Stata, BUGS -> Stan, MATLAB; github for transparent & reproducible research
- presenting research, by conference presentations, peer-reviewed publications, and reports; data visualisation as a powerful way to present the story
- organising scientific meetings/workshops and committee meetings
- awareness of state of the art in statistical methodologies

Teaching & training skills and experience:

- creating and delivering modules and assessment in statistics for undergraduate and postgraduate students, in person and online
- adopting innovative teaching methods; engaging in research on active learning
- supervision of undergraduate and postgraduate students' theses, including collaborations with external partners
- mentorship for undergraduate and postgraduate students as personal tutor

EDUCATION

Master in Education, University Learning and Teaching, Imperial College London.

Thesis Title: *Group Assessment in Mathematics*. Awarded July 2018.

Doctor in Philosophy, Biostatistics, Dublin University, Trinity College, awarded 7/7/2006.

Supervised by Kathleen Bennett.

Thesis Title: *Modelling Compliance with Antihypertensives and Statins in Ireland, using a Prescription Claims Database*.

Master in Science, Statistics, Dublin University, Trinity College, awarded 23/6/2000.

Supervised by Eamonn Mullins. Thesis Title: *Estimating quantile curves, with application to reference curves for Irish birthweights*.

ACADEMIC EXPERIENCE

Lecturer in Medical Biostatistics (above the bar), HRB Clinical Research Facility, National University of Ireland Galway. December 2018 – present. I am engaged in independent and collaborative research on the methodology and application of statistical causal inference in clinical trial and observational studies. I work with large international cohort, case-control and registry studies, primarily in cardiovascular epidemiology. I contributed to the CRFG biostatistics group's response to the COVID-19 pandemic via modelling and predictions of hospital and ICU admissions for hospitals in the west of Ireland. I am co-investigator on an SFI rapid response grant on estimating the effect of non-pharmaceutical interventions on COVID-19 hospital admissions using a mechanistic epidemiological model. From this I have written and submitted papers for publication, including online code and data, as first author, and organised a scientific meeting on statistical methods for COVID-19 on the island of Ireland. I developed and delivered a course on statistical causal inference for PhD and final year Mathematics students.

Teaching Fellow in Statistics, Department of Mathematics, Imperial College London. October 2014-December 2018. My role was exclusively teaching. I lectured several distinct introductory statistics courses for large groups of undergraduate and MSc students from various backgrounds, each course and assessment being tailored to the needs of the group. I developed and delivered courses on Medical Statistics (primarily clinical trials methods), and R & LATEX for MSc in Statistics students. I supported students and provided mentorship as personal tutor. I was supervisor for MSci in Mathematics and MSc in Statistics theses. To enhance my teaching practice, I completed an MEd in University Learning and Teaching.

Senior Statistician, Department of Neonatal Medicine, Imperial College London. September 2012-September 2014. I developed statistical methods to identify unusual performance at hospital level for the UK National Neonatal Audit Programme (NNAP) and other RCPCH audits, and wrote the statistical sections for NNAP annual reports. I contributed to guidance documentation for methodology in clinical audits for HQIP. I collaborated with neonatologists on study design and analysis for clinical trials and observational studies in preterm infants. I engaged in grant writing as co-applicant on major programme grant and individual study grant applications.

Lecturer in Biostatistics, London School of Hygiene and Tropical Medicine. April 2010-September 2012. I provided statistical expertise to an international (UK/US) collaboration on an environmental health study arising from a class action to investigate “probable links” between perfluorooctanoic acid (PFOA) and multiple health outcomes, in part of West Virginia/Ohio where the population was exposed to PFOA in contaminated drinking water. I presented on research at international conferences and was first and contributing author on peer reviewed papers. I taught Basic Statistics, Statistical Methods in Epidemiology, and was tutor and project supervisor to Masters in Public Health students. This developed into an ongoing role of **Deputy Module Organiser**, Basic Statistics for Public Health and Policy, LSHTM Distance Learning. September 2012-present.

Statistician/Epidemiologist, National Perinatal Epidemiology Unit, University of Oxford. Oct 2008-March 2010. I provided statistical expertise with responsibility for co-ordinating a large group of subject experts on a study of relationships between drinking during pregnancy and cognitive and behavioural development of children. This was a statistical genetics study and used a Mendelian Randomisation approach. I presented on the study at conferences and contributed to several peer-reviewed publications as first or joint first author.

Lecturing, Trinity College Dublin, School of Computer Science and Statistics, 2008. I taught Applied Probability to MSISS students (sabbatical cover for Head of Statistics).

Statistician, University College Dublin, School of Public Health and Population Science, Nov 2004-Oct 2008: I worked as an applied statistician on a variety of studies on population health in Ireland. Cohort studies: the Lifeways inter-generational cohort study of children, parents and grandparents, and the Heartwatch study, a longitudinal study of cardiovascular health. Cross-sectional studies: the Survey of Lifestyles, Attitudes and Nutrition (SLAN), Insight 2007 – attitudes to health services in Ireland, and the North Dublin Stroke Study on incidence of stroke in Dublin. Case-control studies: genetic case-control study of cystic fibrosis. I gave presentations at national and international conferences, and contributed to multiple peer-reviewed publications and reports, many as first author. I gave statistics tutorials for undergraduate medical students, lectures for MPH students, and developed and delivered a short course for science PhD students on Statistical Modelling using R.

Lecturing, Trinity College Dublin, School of Radiotherapy & School of Physiotherapy, 2002-2003. I developed and delivered a course in statistics (while a PhD student).

Statistical Research Assistant, Trinity College Dublin, Small Area Health Research Unit, Dept. of Community Health and General Practice. 1998-2000 (while completing MSc).

INDUSTRIAL EXPERIENCE

Statistical Analyst, ABN Amro Leaseholding, IFSC, Dublin 1. 2000-2001. Statistical analysis and prediction of monthly insurance payouts, continued on a consultancy basis during 1st year of PhD.

PUBLICATIONS AND PRESENTATIONS

Journal Publications

- Kennedy SM, O'Driscoll L, Purcell R, Fitz-Simons N, McDermott EW, Hill AD, O'Higgins NJ, Parkinson M, Linehan R, Clynes M. (2003) Prognostic importance of survivin in breast cancer. *Br J Cancer*. **88**:1077-83.
- Hayden PJ, Keogh F, Ni Conghaile M, Carroll M, Crowley M, Fitzsimon N, Gardiner N, Vandenberghe E, O'Riordan J, McCann SR. (2004) A single-centre assessment of long-term quality-of-life status after sibling allogeneic stem cell transplantation for chronic myeloid leukaemia in first chronic phase. *Bone Marrow Transplant*. **34**:545-56.
- Fitz-Simon N, Bennett K, Feely J. (2004) A model for the probability of discontinuation of antihypertensive therapy *British Journal of Clinical Pharmacology* 57(5) 682-3.
- Fitz-Simon N, Bennett K, Feely J (2005). A review of studies of adherence with antihypertensive drugs using prescription databases. *Therapeutics and Clinical Risk Management*, **2**: 15-20.
- Kelleher CC, Lynch JW, Daly L, Harper S, Fitz-Simon N, Bimpeh Y, Daly E, Ulmer H. (2006) The "Americanisation" of migrants: Evidence for the contribution of ethnicity, social deprivation, lifestyle and life-course processes to the mid-20th century Coronary Heart Disease epidemic in the US. *Soc Sci Med*. **63**:465-84.
- Ulmer H, Kelleher CC, Fitz-Simon N, Diem G, Cocin H (2007). Secular trends in cardiovascular risk factors: an age-period-cohort analysis of 698,954 health examinations in 181,350 Austrian men and women. *Journal of Internal Medicine*, **261**:566-576.
- Kelleher C, Fallon U, Fitz-Simon N, Bimpeh Y, Murphy G, Bury G, Murphy AW. (2007) The risk factor profile of grandparents. *Irish Medical Journal*; 100 suppl: 15-19.
- Fitz-Simon N, Johnson H, Hayes C, Mulholland D, Kelleher C. (2007) Predictors of poor self-rated health at area level in the Eastern Regional Health Authority: a multilevel analysis. *Irish Medical Journal*; 100 suppl: 47-49.
- Fitz-Simon N, Shiely F, Corradino D, Friel S, Kelleher C. Predictors of self-reported poor mental health at area level in Ireland: a multilevel analysis of deprivation and social capital indicators. *Irish Medical Journal* ; 100 suppl: 49-52.
- Fitz-Simon N, Fallon U, Sweeney MR, O'Mahony D, Bury G, Murphy AW, Kelleher C for the Lifeways Steering Group. (2007) Mothers' dietary patterns during pregnancy and risk of asthma symptoms in children at 3 years. *Irish Medical Journal*; 100 suppl: 27-32.
- Zuccolo, L* Fitz-Simon N*, Gray R, Ring SM, Davey Smith G, Lewis SJ. (2009) A non-synonymous variant in ADH1B is strongly associated with prenatal alcohol use in a European sample of pregnant women. *Hum Mol Genet*; 18(22):4457-66. * joint 1st author
- Fitz-Simon N, Lewis SJ, Davey Smith G, Zuccolo L, Sayal K, Draper E, Fraser R, Barrow M, Alati R, Ring S, Nutt D, Heron J, Gray R. (2010) Alcohol consumption during pregnancy and children's cognitive development: Mendelian randomisation study using the ADH1B genotype. *Irish Journal of Medical Science*. 179, Suppl 11: S444.
- Fitzpatrick P, Fitz-Simon N, Lonergan M, Collins C, Daly L. (2011) Heartwatch: the effect of a primary care-delivered secondary prevention programme for cardiovascular disease on medication use and risk factor profiles. *Eur J Cardiovasc Prev Rehabil*. 18: 129-135
- Hannon N Sheehan O, Kelly L et al. (2009) Stroke associated with atrial fibrillation--incidence and early outcomes in the north Dublin population stroke study. *Cerebrovasc Dis*; 29(1):43-9.
- Lopez-Espinosa, M.J. Fitz-Simon, N. Bloom, M.S. Calafat, A.M. Fletcher, T. (2012) Comparison between free serum thyroxine levels, measured by analog and dialysis methods,

in the presence of perfluorooctane sulfonate and perfluorooctanoate. *Reprod Toxicol*; 33: 552-555.

Fletcher, T. Lopez-Espinosa, M.J. Armstrong, B. Fitz-Simon, N. Genser, B. Mondal, D. (2011) Patterns of age of puberty in relation to exposure to perfluorooctanoic acid (pfoa) and perfluorooctane sulfonate (pfos) among children living in an area with pfoa-contaminated drinking water *American Journal of Epidemiology*; 173:S43-S43.

Fitz-Simon N, Fletcher T, Luster M, Steenland K, Calafat A, Kato K, Armstrong B. (2013) Reductions in serum lipids with a 4-year decline in serum perfluorooctanoic acid and perfluorooctanesulfonic acid. *Epidemiology*; 24:569-576.

Fitz-Simon N, Fletcher T, Armstrong B. (2013) Understanding uncertainties in a change versus change study. *Epidemiology*; 24:580-581.

Battersby C, Santhakumaran S, Fitz-Simon N, Modi N. (2014) Enteral feed exposures in preterm infants born less than 32 weeks in England. *Archives of Disease in Childhood* **99**: A54.

Battersby C, Mandalia S, Fitz-Simon N, Costello K, Modi N. (2016) A prospective population study of the incidence of severe necrotising enterocolitis in England. *Archives of Disease in Childhood* **101**: A229-A30.

Ferguson J & Fitz-Simon N. Discussion on the meeting “Signs and sizes: understanding and replicating statistical findings”. *J. R. Statist. A* (2020) 183:449-469.

Papers submitted:

Kamiya T, Iglesias-Alvarez A, Ferguson J, Walsh S, Sofonea M, Fitz-Simon N. Estimating time-dependent infectious contact: a multi-strain epidemiological model of SARS-CoV-2 on the island of Ireland. Submitted to Journal of the Royal Society Interface.

doi: <https://doi.org/10.1101/2022.03.25.22272942>

Fitz-Simon N, Ferguson J, Iglesias-Alvarez A, Walsh S, Sofonea M, Kamiya T.

Understanding the role of mask-wearing during COVID-19 on the island of Ireland.

Submitted to JRSSA. **doi:** <https://doi.org/10.1101/2022.03.25.22272946>

Papers in preparation:

Fitz-Simon N and O'Donnell M. Multiple mediation modalities for social class disparities in cardiovascular patients. To submit to epidemiological journal.

Fitz-Simon N and Rodriguez A. Mediation analysis of maternal depression and child neurodevelopment. To submit to child psychology journal.

Fitz-Simon N, O'Donnell M. Characterisation of dietary networks in INTERSTROKE.

Conference and Meeting Presentations

Fitz-Simon N, Bennett K, Feely J. A model for the probability of discontinuation of antihypertensive therapy. Proceedings of the British Pharmacological Society at <http://www.pa2online.org/Vol1Issue4abst217P.html>. December 2002.

Fitz-Simon N, Bennett K, Feely J. A multilevel discrete-time model for competing risks with application to antihypertensive drug use in Ireland. Presented at the 25th Annual Conference of the International Society for Clinical Biostatistics, Leiden, August 15-19, 2004.

Fitz-Simon N, Bennett K, Feely J. A multilevel model for statin discontinuation and switching using a large prescribing database. *Pharmacoepidemiology and Drug Safety*, **13**: S101. (Presented at the International Society for Pharmacoepidemiology Conference, Bordeaux, August 2004).

Fitz-Simon N, Johnson H, Hayes C, Shiely F, Corradino D, Friel S, Kelleher C. Predictors of poor self-reported health at area level in the Eastern Regional Health Authority (ERHA): a multilevel analysis. Presented at the Royal College of Physicians Ireland Faculty of Public Health Summer Scientific Meeting, 2005.

Mulholland D, Fitz-Simon N, Hayes C, Johnson H, Kelleher CC. Is There an Association Between Mortality, Self-Rated Health and GMS Eligibility at District Electoral Division Level in the Eastern Region? Presented at the Royal College of Physicians Ireland Faculty of Public Health Summer Scientific Meeting, 2005.

Fitz-Simon N, Shiely F, Corradino D, Friel S, Kelleher C. Predictors of poor mental health at area level in Ireland: a multilevel analysis of deprivation and social capital indicators. *Journal of Epidemiology and Community Health*. **59** (Suppl 1): A5. Presented at the Society for Social Medicine 49th Annual Scientific Meeting, Glasgow September 14-16 2005.

Fitz-Simon N, Hannon F, Corrigan H Kelleher C. Predictors of physical activity in people with disability or long-term illness in Ireland. Presented at the Irish Social Medicine Meeting, Co Down, 31 March – 2 April 2006.

Corrigan H, Fitz-Simon N, Hannon F, Kelleher C. Predictors of quality of life in people with disability or long-term illness in Ireland. Presented at the Irish Social Medicine Meeting, Co Down, 31 March – 2 April 2006.

Fitz-Simon N, Fallon U, Bury G, Murphy AW, Kelleher C for the Lifeways Steering Group. Predictors of the level of general practice consultations in Lifeways cohort grandparents: a multivariate analysis of socio-demographic and lifestyle risk factors for cardiovascular disease. Presented at the Royal College of Physicians Ireland Faculty of Public Health Summer Scientific Meeting, 2006.

Fitz-Simon N, Fallon U, Sweeney MR, O'Mahony D, Bury G, Murphy AW, Kelleher C for the Lifeways Steering Group. Predictors of asthma in children in Ireland: a multivariate analysis of dietary factors during pregnancy. Presented at the Nutrition Society Meeting (Irish Section), June 14-16 2006.

Fitz-Simon N, Fallon U, O'Mahony D, Bury G, Kelleher C, Murphy AW for the Lifeways Steering Group. Predictors of asthma in children in Ireland: a multivariate analysis of deprivation and social support. *Journal of Epidemiology and Community Health*. **60** (Suppl 1): A12. Presented at the Society for Social Medicine 50th Annual Scientific Meeting, Leeds September 13-15 2006.

Barragry R, Fitzsimon N, Plant B, Chappell SL, Gallagher CG, Daly L, Morgan K, Donnelly SC. Association Analysis of AAT, TNF, ACT, GCSC and GCSR Polymorphisms (SNPs) with Cystic Fibrosis (CF) in an Irish Cohort. American Thoracic Society 2007 International Conference, May 2007.

Fitz-Simon N, O'Mahony D, Boilson A, Craven F, Drummond A, Staines A, Kelleher CC. Satisfaction with General Practice in Insight 2007, the National Consumer Satisfaction Survey: a Graphical Modelling Approach. Presented at the Royal College of Physicians Ireland Faculty of Public Health Winter Scientific Meeting, December 2007.

Fitz-Simon N, Walsh C. Adjusting for unmeasured covariates – an example from patient satisfaction surveys. International Society for Bayesian Analysis annual meeting, Australia, July 21-25 2008.

Fitz-Simon N, Walsh C. Adjusting for unmeasured covariates – A Bayesian Approach. International Biometric Society Conference, Dublin, July 13-18 2008.

Fitz-Simon N, Armstrong B, Fletcher T. The use of instrumental variables to reduce confounding and reverse-causation bias in a cross-sectional study for perfluorooctanoic acid in relation to cholesterol. International Society for Environmental Epidemiology, Barcelona, September 2011.

Fitz-Simon N. Case mix adjustment – what is it, is it useful for the National Neonatal Audit Programme? National Neonatal Audit Programme and Neonatal Data Analysis Unit Collaborators' Meeting, March 2014.

Fitz-Simon N. Assessing evidence from an observational study: C8 and cholesterol in the C8 Science Panel Study. Imperial College London Statistics Seminar, November 2015.

Fitz-Simon N. Assessing evidence from an observational study. Centre for Mathematics of Precision Healthcare Research Meeting, Imperial College London, Monday 22nd January 2018.

Fitz-Simon N. Group Assessment in Mathematics. Imperial College London Mathematics Fellows and Postdocs Seminar, July 2018.

Fitz-Simon N and Rodriguez A. Mediation analysis of maternal depression and child neurodevelopment. European Causal Inference Meeting, Bremen, 27-29 March, 2019.

Fitz-Simon N and Rodriguez A. Mediation analysis of maternal depression and child neurodevelopment. Conference of Applied Statistics in Ireland, Dundalk, 15-17 May 2019.

Fitz-Simon N. Interpreting estimates of mediated effects from studies with attrition: an example from a study of maternal depression and child neurodevelopment. Annual Conference of the Royal Statistical Society, Belfast, 3-5 September 2019.

Fitz-Simon N and O'Donnell M. Multiple mediation modalities for social class disparities in cardiovascular patients. Annual conference of the International Society for Clinical Biostatistics, (Krakow) online, 23-27 August 2020.

Kamiya T, Alvarez-Iglesias A, Ferguson J, Murphy S, Sofonea M, Fitz-Simon N. A multi-strain epidemiological model for SARS-CoV-2 on the island of Ireland. At workshop on statistical approaches to understanding the COVID-19 pandemic on the island of Ireland, NUI Galway, 25th April 2022.

Reports

National Disability Authority (August 2006). Physical activity, health and quality of life among people with disabilities: an analysis of SLAN data. Dublin: NDA.

European Code against Cancer and compliance in an Irish population as assessed from SLAN data. In National Cancer Registry Ireland & The Women's Health Council (February 2006) *Women and Cancer in Ireland 1994-2001*. Dublin: Women's Health Council/National Cancer Registry.

Health Service Executive (September 2007). Insight 07. Health and social services in Ireland – a survey of consumer satisfaction.

Royal College of Paediatrics and Child Health. (August 2013) Annual Report 2012 National Neonatal Audit Programme.

Royal College of Paediatrics and Child Health. (August 2014) Annual Report 2013 National Neonatal Audit Programme.

Royal College of Paediatrics and Child Health. (August 2015) Annual Report 2014 National Neonatal Audit Programme.

PROFESSIONAL DEVELOPMENT

Societies

Member of the Irish Statistical Association.
Fellow of the Royal Statistical Society.
Associate of the Higher Education Academy.

Committees

Chair of the NPEU Staff Committee, Oxford University, 2009.
Postdoc Representative for the Department of Mathematics, Imperial College, 2016-17, winner of the Imperial Postdocs and Fellows Development Centre team award for outstanding contributions to postdoc life in the Department of Mathematics.
Royal Statistical Society Medical Statistics Section committee - Secretary October 2017-2021, I organised and record committee meetings, recorded committee activities, and organise and report on scientific meetings. Discussion Meetings representative 2022.
Royal Statistical Society Florence Nightingale Bicentenary Planning Group, 2019-2020. Planning activities to celebrate the 200th anniversary of Florence Nightingale and her contributions as a statistician.

Conference sessions and scientific meetings organised

RSS Annual Conference invited Session, online, 7-10 September 2020 “Statistics, communication, impact: the Florence Nightingales of today” with speakers Prof Martin Bland, Federica Cocco & John Burn-Murdoch (Financial Times), and Rob Fry (ONS).
RSS invited Session, online, 7-10 September 2020 “Target Trial Emulation” with Prof Bianca de Stavola, Dr Ruth Keogh, and Camille Maringe.
[“Statistical approaches to understanding the covid-19 pandemic on the island of Ireland”](#).
Workshop with 9 speakers at NUI Galway, 25th April 2022, with funding from the ISA and RSS.
Forthcoming: Société Française de Statistique invited session, Lyon 13 June 2022 “Statistics, communication, impact: the Florence Nightingales of today” with speakers Dr Catey Bunce, Dr Victoria Cornelius and Dr Roger Beecham (postponed from 2020).

Good Clinical Practice

Last updated training 16th June 2021

OUTREACH AND ENGAGEMENT ACTIVITIES

Articles and Reports

27th Bradford Hill Memorial Lecture report, *Statslife*, 26th June 2018.
RSS Conference invited session on precision medicine, *Statslife*, 17th September 2019.

29th Bradford Hill Memorial Lecture report, [Statslife](#), 21st July 2020.
30th Bradford Hill Memorial Lecture report, [Statslife](#), 2nd June 2021.

Outreach/engagement talks

Introduction to Statistics for Health Research, presentation for the Community Engaged Scholars Programme, NUI Galway, 30th May 2019.
Data and statistical models to help understand important, real-life questions about causes and effects, Soapbox Science event, Spanish Arch, Galway, 29th June 2019.
A casual chat about causal inference. Famelab, Galway, 19th February 2020.

REFERENCES

Professor Niall Adams
Department of Mathematics
Imperial College London
South Kensington SW7 2AZ
n.adams@imperial.ac.uk

Professor Neena Modi
Department of Neonatal Medicine
Imperial College London
Chelsea and Westminster Hospital
369 Fulham Road
London SW10 9NH
n.modi@imperial.ac.uk

Professor Ben Armstrong
Department of Social and Environmental Health Research
LSHTM
Ben.armstrong@lshtm.ac.uk

Professor David van Dyk
Head of Department
Department of Mathematics
Imperial College London
South Kensington SW7 2AZ
d.van-dyk@imperial.ac.uk